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UNIVERSITY INSTITUTE OF COMPUTING

CASE STUDY

ON

GAMERANK

Esports Tournament and Player Ranking Management System

25CAP-204

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1. INTRODUCTION

GameRank is a database management system developed to manage esports tournaments, players, teams, match results, and player rankings. The project uses MySQL as the database management system and MySQL Workbench as the development tool.

Esports has become increasingly popular, with players participating in different competitive games such as BGMI, Valorant, and CounterStrike 2 (CS2). Managing player information, tournament registrations, match results, and rankings manually can be difficult and timeconsuming.

GameRank provides a structured database to store and manage this information efficiently. It allows users to maintain player records, organize tournaments, register players, record match results, and determine rankings based on points.

The project demonstrates important DBMS concepts, including tables, primary keys, foreign keys, SQL queries, joins, aggregate functions, subqueries, and views.

2. OBJECTIVES

- **To maintain player information, including names, games, teams, countries, and points.**
- **To manage esports teams and their details.**
- **To store tournament information, including tournament names, dates, prize pools, and statuses.**
- **To maintain player registrations for different tournaments.**
- **To record match results, wins, points, and final positions.**

- **To calculate and display player rankings.**
- **To retrieve information using SQL queries.**
- **To reduce data duplication and maintain data consistency using primary keys and foreign keys.**
- **To demonstrate practical implementation of DBMS concepts using MySQL.**

3. PROBLEM STATEMENT

Managing esports tournaments manually can lead to problems such as duplicate records, incorrect player rankings, difficulty tracking tournament registrations, and inefficient management of match results.

Therefore, a database system is required to store and manage esportsrelated information accurately and efficiently.

4. MODULES AND TOOLS AND TECHNOLOGIES USED

A. Modules

1. Player Management Module

This module stores player details such as player ID, player name, game name, team name, country, and total points.

2. Team Management Module

This module maintains team information, including team names, games, and countries.

3. Tournament Management Module

This module stores tournament details such as tournament ID, tournament name, game name, tournament date, prize pool, and tournament status.

4. Registration Management Module

This module records the registration of players in different tournaments. It connects the Players and Tournaments tables through foreign keys.

5. Match Results Management Module

This module stores match statistics, including matches played, matches won, points earned, and final position.

6. Player Ranking Module

This module displays player rankings based on total points. SQL ranking functions can be used to rank players within their respective games.

B. Tools and Technologies Used

1. MySQL

MySQL is a relational database management system used to create databases, tables, relationships, and SQL queries.

2. MySQL Workbench

MySQL Workbench is a graphical tool used to write, execute, and manage SQL commands and view query results.

3. SQL

Structured Query Language (SQL) is used to create tables, insert records, retrieve data, update information, and manage database operations.

Save the dataset as DATA/DATASET.csv relative to the project folder. Expected columns: City, Date, PM2.5, PM10, NO2, SO2, Temperature, Humidity. Column spelling must match the script. The exact results depend on the dataset provided.

Operating System

Windows is used as the operating environment for developing and testing the project.

C. Database Tables

The GameRank project contains five tables:

S. No.	Table Name	Purpose
1	Players	Stores player information
2	Teams	Stores team information
3	Tournaments	Stores tournament details
4	Registrations	Stores tournament registrations
5	MatchResults	Stores match results and scores

5. CODE WITH OUTPUTS

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The project is implemented using SQL commands in MySQL Workbench.

```sql
-- ============================================
-- GAMERANK
-- Esports Tournament and Player Ranking Management System
-- ============================================
CREATE DATABASE GameRank;

USE GameRank;

-- STEP 2: CREATE PLAYERS TABLE

CREATE TABLE Players (
player_id INT PRIMARY KEY AUTO_INCREMENT,
player_name VARCHAR(100) NOT NULL,
game_name VARCHAR(50) NOT NULL,
team_name VARCHAR(100),
country VARCHAR(50),
total_points INT DEFAULT 0
);
```

```sql
-- STEP 3: CREATE TEAMS TABLE

CREATE TABLE Teams (
team_id INT PRIMARY KEY AUTO_INCREMENT,
team_name VARCHAR(100) NOT NULL UNIQUE,
game_name VARCHAR(50),
country VARCHAR(50)
);

-- STEP 4: CREATE TOURNAMENTS TABLE

CREATE TABLE Tournaments (
tournament_id INT PRIMARY KEY AUTO_INCREMENT,
tournament_name VARCHAR(100) NOT NULL,
game_name VARCHAR(50) NOT NULL,
tournament_date DATE,
prize_pool DECIMAL(10,2),
status VARCHAR(20) DEFAULT 'Upcoming'
);
```

```sql
-- STEP 5: CREATE REGISTRATIONS TABLE

CREATE TABLE Registrations (
registration_id INT PRIMARY KEY AUTO_INCREMENT,
player_id INT NOT NULL,
tournament_id INT NOT NULL,
registration_date DATE,
FOREIGN KEY (player_id)
REFERENCES Players(player_id),
FOREIGN KEY (tournament_id)
REFERENCES Tournaments(tournament_id),
UNIQUE (player_id, tournament_id)
);
```

```sql
-- STEP 6: CREATE MATCH RESULTS TABLE

CREATE TABLE MatchResults (
result_id INT PRIMARY KEY AUTO_INCREMENT,
tournament_id INT NOT NULL,
player_id INT NOT NULL,
matches_played INT DEFAULT 0,
matches_won INT DEFAULT 0,
points_earned INT DEFAULT 0,
final_position INT,
FOREIGN KEY (tournament_id)
REFERENCES Tournaments(tournament_id),
FOREIGN KEY (player_id)
REFERENCES Players(player_id)
);
```

```sql
-- STEP 7: INSERT 20 PLAYERS

INSERT INTO Players
(player_name, game_name, team_name, country, total_points)
VALUES
('Aarav Sharma', 'BGMI', 'Team Alpha', 'India', 150),
('Rohan Verma', 'Valorant', 'Team Beta', 'India', 200),
('Priya Singh', 'BGMI', 'Team Alpha', 'India', 180),
('Rahul Kumar', 'Valorant', 'Team Gamma', 'India', 120),
('Ananya Patel', 'BGMI', 'Team Delta', 'India', 220),
('Vikram Singh', 'CS2', 'Team Omega', 'India', 175),
('Neha Gupta', 'Valorant', 'Team Beta', 'India', 240),
('Arjun Mehta', 'BGMI', 'Team Delta', 'India', 190),
('Sneha Reddy', 'CS2', 'Team Sigma', 'India', 210),
('Karan Malhotra', 'Valorant', 'Team Gamma', 'India', 160),
('Ishita Jain', 'BGMI', 'Team Alpha', 'India', 230),
('Aditya Rao', 'CS2', 'Team Omega', 'India', 140),
('Meera Nair', 'Valorant', 'Team Beta', 'India', 195),
('Kabir Das', 'BGMI', 'Team Delta', 'India', 205),
('Pooja Shah', 'CS2', 'Team Sigma', 'India', 185),
('Dev Kapoor', 'Valorant', 'Team Gamma', 'India', 250),
('Simran Kaur', 'BGMI', 'Team Alpha', 'India', 170),
('Yash Thakur', 'CS2', 'Team Omega', 'India', 225),
('Tanya Bose', 'Valorant', 'Team Beta', 'India', 155),
('Manav Joshi', 'BGMI', 'Team Delta', 'India', 215);

-- STEP 8: INSERT 10 TEAMS

INSERT INTO Teams
(team_name, game_name, country)
VALUES
('Team Alpha', 'BGMI', 'India'),
('Team Beta', 'Valorant', 'India'),
('Team Gamma', 'Valorant', 'India'),
('Team Delta', 'BGMI', 'India'),
('Team Omega', 'CS2', 'India'),
('Team Sigma', 'CS2', 'India'),
('Team Phoenix', 'BGMI', 'India'),
('Team Titans', 'Valorant', 'India'),
('Team Warriors', 'CS2', 'India'),
('Team Legends', 'BGMI', 'India');
```

```sql
-- STEP 9: INSERT 10 TOURNAMENTS

INSERT INTO Tournaments
(tournament_name, game_name, tournament_date, prize_pool, status)
VALUES
('BGMI Championship', 'BGMI', '2026-11-15', 50000, 'Upcoming'),
('Valorant Masters', 'Valorant', '2026-12-01', 75000, 'Upcoming'),
('Winter Esports Cup', 'BGMI', '2026-10-10', 30000, 'Completed'),
('CS2 Pro League', 'CS2', '2026-11-20', 60000, 'Upcoming'),
('National Gaming Cup', 'BGMI', '2026-12-10', 45000, 'Upcoming'),
('Valorant Elite Cup', 'Valorant', '2026-12-15', 80000, 'Upcoming'),
('Summer Gaming League', 'CS2', '2026-09-15', 35000, 'Completed'),
('India Esports Open', 'BGMI', '2026-11-25', 55000, 'Upcoming'),
('Champions Arena', 'Valorant', '2026-10-05', 90000, 'Completed'),
('Pro Gamers Tournament', 'CS2', '2026-12-20', 70000, 'Upcoming');

```

```sql
-- STEP 10: INSERT 30 REGISTRATIONS

INSERT INTO Registrations
(player_id, tournament_id, registration_date)
VALUES
(1, 1, '2026-10-20'),
(2, 2, '2026-10-20'),
(3, 1, '2026-10-21'),
(4, 2, '2026-10-21'),
(5, 5, '2026-10-22'),
(6, 4, '2026-10-22'),
(7, 2, '2026-10-23'),
(8, 1, '2026-10-23'),
(9, 4, '2026-10-23'),
(10, 6, '2026-10-24'),
(11, 5, '2026-10-24'),
(12, 7, '2026-09-01'),
(13, 6, '2026-10-24'),
(14, 8, '2026-10-24'),
(15, 4, '2026-10-24'),
(16, 7, '2026-09-20'),
(17, 1, '2026-10-25'),
(18, 10, '2026-10-25'),
(19, 6, '2026-10-25'),
(20, 8, '2026-10-25'),
(1, 3, '2026-09-20'),
(3, 3, '2026-09-21'),
(6, 7, '2026-09-02'),
(9, 7, '2026-09-03'),
(12, 4, '2026-10-26'),
(15, 7, '2026-09-04'),
(2, 6, '2026-10-26'),
(7, 9, '2026-09-15'),
(11, 8, '2026-10-26'),
(18, 4, '2026-10-27');
```

```sql
-- STEP 11: INSERT 30 MATCH RESULTS

INSERT INTO MatchResults
(tournament_id, player_id, matches_played,
matches_won, points_earned, final_position)
VALUES
(3, 1, 5, 3, 100, 2),
(2, 2, 6, 4, 200, 1),
(3, 3, 5, 4, 150, 1),
(2, 4, 6, 2, 80, 3),
(5, 5, 4, 3, 120, 2),
(4, 6, 7, 5, 180, 1),
(2, 7, 6, 5, 220, 1),
(1, 8, 5, 3, 130, 3),
(4, 9, 7, 4, 160, 2),
(6, 10, 5, 3, 140, 3),
(5, 11, 4, 4, 190, 1),
(7, 12, 6, 2, 90, 4),
(6, 13, 5, 3, 170, 2),
(8, 14, 4, 3, 150, 2),
(4, 15, 7, 3, 110, 3),
(9, 16, 6, 5, 250, 1),
(1, 17, 5, 2, 100, 4),
(10, 18, 6, 4, 210, 1),
```

```
    (6, 19, 5, 3, 135, 3),
    (8, 20, 4, 2, 95, 4),
    (3, 1, 5, 3, 110, 2),
    (3, 3, 5, 4, 160, 1),
    (7, 6, 6, 4, 175, 2),
    (7, 9, 6, 5, 200, 1),
    (4, 12, 7, 3, 125, 3),
    (7, 15, 6, 2, 85, 4),
    (6, 2, 5, 4, 190, 1),
    (9, 7, 6, 4, 205, 2),
    (8, 11, 4, 3, 145, 2),
    (4, 18, 7, 5, 230, 1);

    -- STEP 12: DISPLAY ALL TABLES

SHOW TABLES;

    -- STEP 13: DISPLAY ALL PLAYERS

SELECT * FROM Players;

    -- STEP 14: DISPLAY ALL TEAMS

SELECT * FROM Teams;
```

```
    -- STEP 15: DISPLAY ALL TOURNAMENTS

SELECT * FROM Tournaments;

    -- STEP 16: DISPLAY ALL REGISTRATIONS

SELECT * FROM Registrations;

    -- STEP 17: DISPLAY ALL MATCH RESULTS

SELECT * FROM MatchResults;

    -- STEP 18: DISPLAY PLAYERS BY POINTS

SELECT player_name, game_name, total_points
FROM Players
ORDER BY total_points DESC;

    -- STEP 19: DISPLAY PLAYERS WITH MORE THAN 180 POINTS

SELECT player_name, game_name, total_points
FROM Players
WHERE total_points > 180;

```

```
    -- STEP 20: COUNT PLAYERS IN EACH GAME

SELECT game_name, COUNT(*) AS total_players
FROM Players
GROUP BY game_name;

    -- STEP 21: COUNT REGISTRATIONS IN EACH TOURNAMENT

SELECT T.tournament_name,
COUNT(R.registration_id) AS total_registrations
FROM Tournaments T
LEFT JOIN Registrations R
ON T.tournament_id = R.tournament_id
GROUP BY T.tournament_id, T.tournament_name;
```

```
    -- STEP 22: DISPLAY PLAYER TOURNAMENT REGISTRATIONS

SELECT P.player_name,
T.tournament_name,
R.registration_date
FROM Registrations R
JOIN Players P
ON R.player_id = P.player_id
JOIN Tournaments T
ON R.tournament_id = T.tournament_id;

    -- STEP 23: DISPLAY MATCH RESULTS

SELECT P.player_name,
T.tournament_name,
M.matches_played,
M.matches_won,
M.points_earned,
M.final_position
FROM MatchResults M
JOIN Players P
ON M.player_id = P.player_id
JOIN Tournaments T
ON M.tournament_id = T.tournament_id;
```

```sql
-- STEP 24: FIND THE HIGHEST-SCORING PLAYER

SELECT player_name, game_name, total_points
FROM Players
ORDER BY total_points DESC
LIMIT 1;

-- STEP 25: FIND AVERAGE PLAYER POINTS

SELECT AVG(total_points) AS average_points
FROM Players;

-- STEP 26: FIND PLAYERS WITH ABOVE-AVERAGE POINTS

SELECT player_name, total_points
FROM Players
WHERE total_points > (
SELECT AVG(total_points)
FROM Players
);
```

```sql
-- STEP 27: CREATE PLAYER RANKINGS VIEW

CREATE VIEW PlayerRankings AS
SELECT player_id,
player_name,
game_name,
team_name,
total_points,
RANK() OVER (
PARTITION BY game_name
ORDER BY total_points DESC
) AS player_rank
FROM Players;

-- STEP 28: DISPLAY PLAYER RANKINGS

SELECT * FROM PlayerRankings;

-- STEP 29: CREATE INDEX

CREATE INDEX idx_game_points
ON Players(game_name, total_points);
```

```sql
-- STEP 30: VERIFY RECORD COUNTS

SELECT COUNT(*) AS total_players FROM Players;

SELECT COUNT(*) AS total_teams FROM Teams;

SELECT COUNT(*) AS total_tournaments FROM Tournaments;

SELECT COUNT(*) AS total_registrations FROM Registrations;

SELECT COUNT(*) AS total_match_results FROM MatchResults;
```

OUTPUTS:

Tables_in_gamerank

- matchresults
- players
- registrations
- teams
- tournaments

player_id	player_name	game_name	team_name	country	total_points
1	Aarav Sharma	BGMI	Team Alpha	India	150
2	Rohan Verma	Valorant	Team Beta	India	200
3	Priya Singh	BGMI	Team Alpha	India	180
4	Rahul Kumar	Valorant	Team Gamma	India	120
5	Ananya Patel	BGMI	Team Delta	India	220
6	Vikram Singh	CS2	Team Omega	India	175
7	Neha Gupta	Valorant	Team Beta	India	240
8	Arjun Mehta	BGMI	Team Delta	India	190
9	Sneha Reddy	CS2	Team Sigma	India	210
10	Karan Malhotra	Valorant	Team Gamma	India	160
11	Ishita Jain	BGMI	Team Alpha	India	230
12	Aditya Rao	CS2	Team Omega	India	140
13	Meera Nair	Valorant	Team Beta	India	195
14	Kabir Das	BGMI	Team Delta	India	205
15	Pooja Shah	CS2	Team Sigma	India	185
16	Dev Kapoor	Valorant	Team Gamma	India	250
17	Simran Kaur	BGMI	Team Alpha	India	170
18	Yash Thakur	CS2	Team Omega	India	225
19	Tanya Bose	Valorant	Team Beta	India	155
20	Manav Joshi	BGMI	Team Delta	India	215

team_id	team_name	game_name	country
1	Team Alpha	BGMI	India
2	Team Beta	Valorant	India
3	Team Gamma	Valorant	India
4	Team Delta	BGMI	India
5	Team Omega	CS2	India
6	Team Sigma	CS2	India
7	Team Phoenix	BGMI	India
8	Team Titans	Valorant	India
9	Team Warriors	CS2	India
10	Team Legends	BGMI	India

tournament_id	tournament_name	game_name	tournament_date	prize_pool	status
1	BGMI Championship	BGMI	11/15/2026	50000.00	Upcoming
2	Valorant Masters	Valorant	12/01/2026	75000.00	Upcoming
3	Winter Esports Cup	BGMI	10/10/2026	30000.00	Completed
4	CS2 Pro League	CS2	11/20/2026	60000.00	Upcoming
5	National Gaming Cup	BGMI	12/10/2026	45000.00	Upcoming
6	Valorant Elite Cup	Valorant	12/15/2026	80000.00	Upcoming
7	Summer Gaming League	CS2	09/15/2026	35000.00	Completed
8	India Esports Open	BGMI	11/25/2026	55000.00	Upcoming
9	Champions Arena	Valorant	10/05/2026	90000.00	Completed
10	Pro Gamers Tournament	CS2	12/20/2026	70000.00	Upcoming

registration_id	player_id	tournament_id	registration_date
1	1	1	10/20/2026
2	2	2	10/20/2026
3	3	1	10/21/2026
4	4	2	10/21/2026
5	5	5	10/22/2026
6	6	4	10/22/2026
7	7	2	10/23/2026
8	8	1	10/23/2026
9	9	4	10/23/2026
10	10	6	10/24/2026
11	11	5	10/24/2026
12	12	7	09/01/2026
13	13	6	10/24/2026
14	14	8	10/24/2026
15	15	4	10/24/2026
16	16	7	09/20/2026
17	17	1	10/25/2026
18	18	10	10/25/2026
19	19	6	10/25/2026
20	20	8	10/25/2026
21	1	3	09/20/2026
22	3	3	09/21/2026
23	6	7	09/02/2026
24	9	7	09/03/2026
25	12	4	10/26/2026
26	15	7	09/04/2026
27	2	6	10/26/2026
28	7	9	09/15/2026
29	11	8	10/26/2026
30	18	4	10/27/2026

result_id	tournament_id	player_id	matches_played	matches_won	points_earned	final_position
1	3	1	5	3	100	2
2	2	2	6	4	200	1
3	3	3	5	4	150	1
4	2	4	6	2	80	3
5	5	5	4	3	120	2
6	4	6	7	5	180	1
7	2	7	6	5	220	1
8	1	8	5	3	130	3
9	4	9	7	4	160	2
10	6	10	5	3	140	3
11	5	11	4	4	190	1
12	7	12	6	2	90	4
13	6	13	5	3	170	2
14	8	14	4	3	150	2
15	4	15	7	3	110	3
16	9	16	6	5	250	1
17	1	17	5	2	100	4

18	10	18	6	4	210	1
19	6	19	5	3	135	3
20	8	20	4	2	95	4
21	3	1	5	3	110	2
22	3	3	5	4	160	1
23	7	6	6	4	175	2
24	7	9	6	5	200	1
25	4	12	7	3	125	3
26	7	15	6	2	85	4
27	6	2	5	4	190	1
28	9	7	6	4	205	2
29	8	11	4	3	145	2
30	4	18	7	5	230	1

player_name	game_name	total_points
Dev Kapoor	Valorant	250
Neha Gupta	Valorant	240
Ishita Jain	BGMI	230
Yash Thakur	CS2	225
Ananya Patel	BGMI	220
Manav Joshi	BGMI	215
Sneha Reddy	CS2	210
Kabir Das	BGMI	205
Rohan Verma	Valorant	200
Meera Nair	Valorant	195
Arjun Mehta	BGMI	190
Pooja Shah	CS2	185
Priya Singh	BGMI	180
Vikram Singh	CS2	175
Simran Kaur	BGMI	170
Karan Malhotra	Valorant	160
Tanya Bose	Valorant	155

Aarav Sharma	BGMI	150
Aditya Rao	CS2	140
Rahul Kumar	Valorant	120

player_name	game_name	total_points
Rohan Verma	Valorant	200
Ananya Patel	BGMI	220
Neha Gupta	Valorant	240
Arjun Mehta	BGMI	190
Sneha Reddy	CS2	210
Ishita Jain	BGMI	230
Meera Nair	Valorant	195
Kabir Das	BGMI	205
Pooja Shah	CS2	185
Dev Kapoor	Valorant	250
Yash Thakur	CS2	225
Manav Joshi	BGMI	215

game_name	total_players
BGMI	8
Valorant	7
CS2	5

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tournament_name	total_registrations
BGMI Championship	4
Valorant Masters	3
Winter Esports Cup	2
CS2 Pro League	5
National Gaming Cup	2
Valorant Elite Cup	4
Summer Gaming League	5
India Esports Open	3
Champions Arena	1
Pro Gamers Tournament	1

player_name	tournament_name	registration_date
Aarav Sharma	BGMI Championship	10/20/2026
Priya Singh	BGMI Championship	10/21/2026
Arjun Mehta	BGMI Championship	10/23/2026
Simran Kaur	BGMI Championship	10/25/2026
Rohan Verma	Valorant Masters	10/20/2026
Rahul Kumar	Valorant Masters	10/21/2026
Neha Gupta	Valorant Masters	10/23/2026
Aarav Sharma	Winter Esports Cup	09/20/2026
Priya Singh	Winter Esports Cup	09/21/2026
Vikram Singh	CS2 Pro League	10/22/2026
Sneha Reddy	CS2 Pro League	10/23/2026
Pooja Shah	CS2 Pro League	10/24/2026
Aditya Rao	CS2 Pro League	10/26/2026
Yash Thakur	CS2 Pro League	10/27/2026
Ananya Patel	National Gaming Cup	10/22/2026
Ishita Jain	National Gaming Cup	10/24/2026
Karan Malhotra	Valorant Elite Cup	10/24/2026
Meera Nair	Valorant Elite Cup	10/24/2026
Tanya Bose	Valorant Elite Cup	10/25/2026
Rohan Verma	Valorant Elite Cup	10/26/2026
Aditya Rao	Summer Gaming League	09/01/2026
Dev Kapoor	Summer Gaming League	09/20/2026
Vikram Singh	Summer Gaming League	09/02/2026
Sneha Reddy	Summer Gaming League	09/03/2026
Pooja Shah	Summer Gaming League	09/04/2026
Kabir Das	India Esports Open	10/24/2026
Manav Joshi	India Esports Open	10/25/2026
Ishita Jain	India Esports Open	10/26/2026
Neha Gupta	Champions Arena	09/15/2026
Yash Thakur	Pro Gamers Tournament	10/25/2026

player_name	tournament_name	matches_played	matches_won	points_earned	final_position
Arjun Mehta	BGMI Championship	5	3	130	3
Simran Kaur	BGMI Championship	5	2	100	4
Rohan Verma	Valorant Masters	6	4	200	1
Rahul Kumar	Valorant Masters	6	2	80	3
Neha Gupta	Valorant Masters	6	5	220	1
Aarav Sharma	Winter Esports Cup	5	3	100	2
Priya Singh	Winter Esports Cup	5	4	150	1
Aarav Sharma	Winter Esports Cup	5	3	110	2
Priya Singh	Winter Esports Cup	5	4	160	1
Vikram Singh	CS2 Pro League	7	5	180	1
Sneha Reddy	CS2 Pro League	7	4	160	2
Pooja Shah	CS2 Pro League	7	3	110	3
Aditya Rao	CS2 Pro League	7	3	125	3
Yash Thakur	CS2 Pro League	7	5	230	1
Ananya Patel	National Gaming Cup	4	3	120	2
Ishita Jain	National Gaming Cup	4	4	190	1
Karan Malhotra	Valorant Elite Cup	5	3	140	3
Meera Nair	Valorant Elite Cup	5	3	170	2
Tanya Bose	Valorant Elite Cup	5	3	135	3
Rohan Verma	Valorant Elite Cup	5	4	190	1
Aditya Rao	Summer Gaming League	6	2	90	4
Vikram Singh	Summer Gaming League	6	4	175	2
Sneha Reddy	Summer Gaming League	6	5	200	1
Pooja Shah	Summer Gaming League	6	2	85	4
Kabir Das	India Esports Open	4	3	150	2
Manav Joshi	India Esports Open	4	2	95	4
Ishita Jain	India Esports Open	4	3	145	2
Dev Kapoor	Champions Arena	6	5	250	1
Neha Gupta	Champions Arena	6	4	205	2
Yash Thakur	Pro Gamers Tournament	6	4	210	1

player_name	game_name	total_points
Dev Kapoor	Valorant	250

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player_name	total_points
Rohan Verma	200
Ananya Patel	220
Neha Gupta	240
Sneha Reddy	210
Ishita Jain	230
Meera Nair	195
Kabir Das	205
Dev Kapoor	250
Yash Thakur	225
Manav Joshi	215

average_points
190.7500

player_id	player_name	game_name	team_name	total_points	player_rank
11	Ishita Jain	BGMI	Team Alpha	230	1
5	Ananya Patel	BGMI	Team Delta	220	2
20	Manav Joshi	BGMI	Team Delta	215	3
14	Kabir Das	BGMI	Team Delta	205	4
8	Arjun Mehta	BGMI	Team Delta	190	5
3	Priya Singh	BGMI	Team Alpha	180	6
17	Simran Kaur	BGMI	Team Alpha	170	7
1	Aarav Sharma	BGMI	Team Alpha	150	8
18	Yash Thakur	CS2	Team Omega	225	1
9	Sneha Reddy	CS2	Team Sigma	210	2
15	Pooja Shah	CS2	Team Sigma	185	3
6	Vikram Singh	CS2	Team Omega	175	4
12	Aditya Rao	CS2	Team Omega	140	5
16	Dev Kapoor	Valorant	Team Gamma	250	1
7	Neha Gupta	Valorant	Team Beta	240	2
2	Rohan Verma	Valorant	Team Beta	200	3
13	Meera Nair	Valorant	Team Beta	195	4
10	Karan Malhotra	Valorant	Team Gamma	160	5
19	Tanya Bose	Valorant	Team Beta	155	6
4	Rahul Kumar	Valorant	Team Gamma	120	7

total_players	total_teams	total_tournaments	total_registrations	total_match_results
20	10	10	30	30

6. CONCLUSION

The GameRank project successfully demonstrates the application of database management concepts to esports tournament and player ranking management.

The system provides an organized way to maintain player details, teams, tournaments, registrations, and match results. It uses relational tables, primary keys, foreign keys, and SQL queries to maintain relationships and retrieve useful information.

The project also demonstrates the use of joins, aggregate functions, subqueries, and views to generate tournament statistics and player rankings.

Through this project, practical knowledge of MySQL and relational database management has been developed. The system can be extended in the future by adding features such as team-based tournament registrations, automatic point calculations, user authentication, and a graphical interface.

In conclusion, GameRank is a useful DBMS mini project that demonstrates how SQL can be applied to manage esports data efficiently and systematically.

GIT HUB - https://github.com/Mayank-Sharma7/GameRank-Esports-Tournament-Management